

Zachary Neagle

RoboBoston 2026 - Electromechanical Technician · Robotics Transition

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Hands-on multidisciplinary technician moving deliberately into robotics and electromechanical systems. Background spans audio engineering and routed-system troubleshooting, mechanical and automotive diagnostics, metal fabrication and welding, sensor-driven interactive systems, electrical study, construction, equipment operation, and field problem solving. Strongest recurring skill is tracing consequential signal, control, and failure paths across interconnected systems. Seeking robotics, integration, test, reliability, prototype-build, field-support, and electromechanical roles where existing physical-systems competence can contribute immediately while deeper robotics expertise is developed professionally.

Technical foundation

Signal paths / systems troubleshooting

Years of self-directed audio engineering developed mental modeling of signal paths, gain stages, processing chains, control points, dependencies, downstream effects, and methodical fault isolation.

Sensors / controls / interactive systems

Troubleshoot systems involving RFID and other sensors, PLC-monitored elements, DMX lighting, Arduino hardware, programmable LEDs, audio, mechanical components, and computer controls. Used PLC monitoring interfaces to observe system states and test sensors; programming and ladder-logic changes were handled by software personnel.

Mechanical systems / diagnostics

Long-term self-directed hands-on diagnosis and repair of gasoline- and diesel-powered vehicles, dirt bikes, ATVs, and small engines; familiarity with fuel, ignition, starting/charging, cooling, lubrication, wiring, sensors, and controls.

Fabrication / build execution

GMAW and SMAW welding, oxy-fuel cutting, brake, shear, punch, drill press, structural fabrication, fitting, rigging, machinery, field installation, material handling, and physical build sequencing.

Electrical foundation

Completed Martin Electrical School's Journeyman One program in May 2026. Studied electrical theory, code, safety, and trade fundamentals; one of two students in the class to score 100 on the midterm examination.

Selected relevant experience

Venue Technician

Level99

Natick, MA - Jul 2024-Jun 2025

- Maintained and troubleshooted interactive systems integrating RFID and other sensors, PLC-monitored elements, DMX lighting, Arduino hardware, programmable LEDs, audio, mechanical components, and computer-based controls.
- Used PLC monitoring interfaces primarily to observe system states and test sensors while supporting broad venue fault diagnosis, preventive maintenance, and physical repair.
- Worked at the boundary between software-controlled behavior and real physical mechanisms in a high-use environment.

Project-Based Construction & Technical Installations

Tradesmen International / Wide Effect Talent Solutions / Suncat Solutions

National / MA / TX / TN / FL - 2022-2025

- Completed traveling assignments spanning manufacturing construction, commercial millwork, high-end remodeling, interactive entertainment installation, and technical/scenic builds.
- Built clean-room structures for PCB manufacturing at the Starlink facility in Bastrop, Texas.
- Completed Hyper Bowling installations and Escape Game technical/scenic builds; adapted quickly to unfamiliar crews, systems, sites, and production constraints.

Welder / Fabricator

Superior Rail & Iron Works

East Bridgewater, MA - May 2017-Nov 2019

- Fabricated stairs and rails for steel-frame buildings using GMAW/SMAW welding, oxy-fuel cutting, brake, shear, punch, and drill-press operations.
- Supported fitting, finishing, rigging, shipping, loading, deliveries, and installation logistics.

Carpenter

Absolute Home Improvement

Lakeville, MA - Apr 2014-May 2022

- Developed a long-term foundation in complete residential remodeling, including framing, windows, siding, roofing, decks, flooring, trim, painting, repair, and customer-facing work in occupied homes.
- Built practical judgment around sequencing, access, measurement, material behavior, repairability, and field adaptation.

Education

Martin Electrical School - Norwood, MA

Journeyman One - Certificate of First-Year Completion, May 2026.

Southeastern Regional Vocational Technical High School - Easton, MA

Multi-Process Welding, Metal Fabrication & Machine Technology, May 2013.

Selected hands-on tooling

Electrical / electronics

Multimeter, oscilloscope, bench power supply, soldering station, wire strippers/crimpers, heat-shrink work.

Fabrication / shop

MIG, TIG, and stick welding; oxy-fuel torch; angle grinder; drill press; brake and shear; ironworker/punch machine.

Mechanical / diagnostics

Torque wrench, compression tester, OBD-II scan tool, timing light, calipers and micrometer, dial indicator.

Controls / signal

DMX lighting console, Arduino, Raspberry Pi, PLC hardware, RFID readers, digital mixing console.

Software

VS Code, Git/GitHub, Linux shell, Python, JavaScript/HTML/CSS, Node.js/npm, AI coding tools including Claude Code and Codex.

Additional qualifications

OSHA 10 - Qualified rigger / signal person - Telehandler, skid loader, self-propelled and boom-supported work-platform experience - Valid driver's license - Reliable transportation - Available for travel

What I'm targeting

Robotics / mechatronics hardware technician, prototype and integration/test, field robotics and deployment, reliability/root-cause work, and manufacturing/test engineering support where hands-on execution can grow into increasing engineering ownership.